AI-Powered Multi-Map Framework for Keratoconus Classification in Bangladeshi Eye Images

My Undergraduate Thesis | CSE4100 & CSE4250 | Ahsanullah University of Science and Technology (AUST)
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What is This Project About?

The Problem

Keratoconus is an eye disease where the cornea (the clear front part of your eye) gets thinner and bulges outward into a cone shape. If not caught early, it can cause serious vision loss and may require a cornea transplant.

The challenge? Diagnosing it early is hard. Doctors use a special machine called **Pentacam** that takes 7 different pictures (maps) of your cornea. But reading these maps requires years of specialized training — and in Bangladesh, most cornea specialists are only in big cities like Dhaka. People in rural areas often get diagnosed too late.

What I Built

I created an **AI system** that automatically reads all 7 Pentacam maps together and tells whether an eye is:

- **Normal** — healthy eye
- **Suspect** — early signs of keratoconus (needs monitoring)
- **Keratoconus** — confirmed disease (needs treatment)

Why It Matters

- **First study** to test keratoconus AI on **Bangladeshi patients** (South Asian population)
- Uses **all 7 maps together** — most previous work only used 1 or 2
- Achieves **93.3% sensitivity** — catches almost all real cases
- Works on **standard hospital equipment** — no extra hardware needed

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How Does It Work? (Simple Explanation)

Imagine you're trying to figure out if a fruit is ripe. You could look at its color, smell it, or squeeze it. Each clue helps, but using ALL of them together gives you the best answer.

That's what this system does with your eye:

Map	What It Shows	Why It Matters
SAG_A	Front curve of cornea	Keratoconus makes this steeper
SAG_P	Back curve of cornea	Shows early changes

Table 1 – continued

Map	What It Shows	Why It Matters
ELV__A	Front elevation	Detects bulging
ELV__P	Back elevation	Catches subtle changes
CT__A	Cornea thickness	Keratoconus thins the cornea
EC__A	Front ectasia index	Measures how much the front is warped
EC__P	Back ectasia index	Measures how much the back is warped

The AI reads all 7 maps together — like a doctor who never gets tired and has seen thousands of eyes.

Results

What We Achieved

Test	Accuracy	What It Means
International Test (Brazilian data)	**80.2%**	Works on data from another country
Bangladeshi Test (local patients)	**74.7%**	Works on our own population
Binary Screening (Normal vs. Abnormal)	**84.5%**	Good for first-level screening
Sensitivity (catching real cases)	**93.3%**	Misses very few sick patients

Why the Bangladeshi Number is Lower

South Asian corneas have slightly different shapes due to genetics, environment, and lifestyle factors (like eye rubbing from allergies). This is exactly why we needed this study — most AI systems are trained on Western or Latin American eyes and don't work as well here.

The good news: 93.3% sensitivity means the system catches almost all real keratoconus cases, making it safe for screening.

Project Structure

```
keratoconus-ai/
│
├── README.md                    # This file — you're reading it!
├── requirements.txt             # Python packages needed
├── thesis.pdf                   # Full thesis document
│
├── data/
│   ├── international/           # Brazilian dataset (from Kaggle)
```

```
│   │   ├── train/
│   │   │   ├── Normal/
│   │   │   ├── Suspect/
│   │   │   └── Keratoconus/
│   │   ├── test/
│   │   │   ├── Normal/
│   │   │   ├── Suspect/
│   │   │   └── Keratoconus/
│   │   └── val/
│   │       ├── Normal/
│   │       ├── Suspect/
│   │       └── Keratoconus/
│   │
│   └── bangladeshi/                  # Local data (NOT included — see note below)
│       └── README.md                 # Instructions for NIOH collaborators
│
├── src/
│   ├── config.py                     # All settings in one place
│   ├── preprocess.py                 # Clean and prepare images
│   ├── extract_features.py           # Extract features using DenseNet-121
│   ├── train_classifier.py           # Train the SVM classifier
│   ├── evaluate.py                   # Test the model and get results
│   ├── gradcam_visualize.py          # Create heatmaps to explain predictions
│   └── utils.py                      # Helper functions
│
├── notebooks/
│   ├── 01_data_exploration.ipynb      # Look at the data
│   ├── 02_single_map_baseline.ipynb   # Test each map alone
│   ├── 03_multi_map_fusion.ipynb      # Combine all 7 maps
│   ├── 04_model_comparison.ipynb      # Compare CNN backbones
│   └── 05_interpretability.ipynb      # Grad-CAM visualizations
│
├── models/                           # Saved trained models
│   └── best_model.pkl
│
├── results/                          # Output files
│   ├── confusion_matrices/
│   ├── roc_curves/
│   ├── gradcam_heatmaps/
│   └── metrics.json
│
└── tests/                            # Unit tests
    └── test_preprocessing.py
```

Dataset Information

International Dataset (Included — From Kaggle)

We use the publicly available **Brazilian Keratoconus Dataset** from Kaggle for training and initial testing:

- **Source:** https://www.kaggle.com/datasets/elmehdi12/keratoconus-detection
- **Patients:** 573 eyes from Brazil
- **Classes:** Normal (200), Suspect (173), Keratoconus (200)
- **Maps per eye:** 7 (SAG_A, SAG_P, ELV_A, ELV_P, CT_A, EC_A, EC_P)
- **Format:** Pre-segmented images

Bangladeshi Dataset (NOT Included — Legal Restrictions)

- **Patients:** 142 Bangladeshi patients (284 eyes) from NIOH, Dhaka
- **Collection:** Printed Pentacam reports manually processed
- **Why not shared:** Patient privacy and hospital data protection policies
- **Labels:** Verified by Dr. S M Enamul Haque (Associate Professor, NIOH)

> **Note for Researchers:** If you're affiliated with a medical institution and want to collaborate, contact me or my supervisor. We can discuss proper data sharing agreements.

How to Run This Project

Step 1: What You Need (Prerequisites)

Before starting, make sure you have:

Requirement	Version	Why You Need It
Python	3.10 or higher	The programming language
pip	Latest	To install packages
Git	Latest	To download this code
8GB+ RAM	—	For processing images
GPU (optional)	NVIDIA with CUDA	Speeds up training a lot

Don't have Python? Download from python.org and make sure to check "Add Python to PATH" during installation.

Step 2: Download the Code

Open your terminal (Command Prompt on Windows, Terminal on Mac/Linux) and type:

```
# Clone this repository
git clone https://github.com/yourusername/keratoconus-ai.git

# Go into the folder
cd keratoconus-ai
```

Step 3: Set Up Virtual Environment (Recommended)

This keeps your project packages separate from other projects:

```
# Create virtual environment
python -m venv venv

# Activate it
# On Windows:
venv\Scripts\activate
# On Mac/Linux:
source venv/bin/activate
```

You'll see `(venv)` at the start of your command line — that means it's working.

Step 4: Install Required Packages

```
pip install -r requirements.txt
```

This downloads all the Python libraries we need (PyTorch, Scikit-learn, OpenCV, etc.). It might take 5-10 minutes depending on your internet.

Step 5: Download the Dataset

1. Go to https://www.kaggle.com/datasets/elmehdi12/keratoconus-detection
2. Click **Download** (you need a free Kaggle account)
3. Unzip the file
4. Place the folders inside `data/international/` so it looks like:

```
data/international/
├── train/
│   ├── Normal/
│   ├── Suspect/
│   └── Keratoconus/
├── test/
│   ├── Normal/
│   ├── Suspect/
│   └── Keratoconus/
└── val/
    ├── Normal/
    ├── Suspect/
    └── Keratoconus/
```

Step 6: Run the Full Pipeline

We provide a simple script that runs everything:

```
# Run the complete pipeline
python run_full_pipeline.py
```

This will:

1. Preprocess all images

2. Extract features using DenseNet-121
3. Train the SVM classifier
4. Evaluate on test sets
5. Generate visualizations

Expected time: 30-60 minutes on CPU, 10-15 minutes on GPU.

Step 7: See Your Results

After running, check the `results/` folder:

```
results/
├── confusion_matrices/         # See how well the model classified each class
├── roc_curves/                 # How good is the model at distinguishing classes?
├── gradcam_heatmaps/           # Where is the AI looking? (explainability)
└── metrics.json                # All numbers (accuracy, precision, recall, F1)
```

Running Individual Parts (For Learning)

Want to understand each step? Run them one by one:

1. Preprocess Images

```
python src/preprocess.py
```

What it does: Crops the cornea from raw images, resizes to 380×380, converts to grayscale, and applies data augmentation (flip, rotate, brightness) for training images.

2. Extract Features

```
python src/extract_features.py
```

What it does: Runs each of the 7 maps through DenseNet-121 (a pre-trained neural network) to convert images into 1,024 numbers per map. These numbers capture the "essence" of what the image shows.

3. Train Classifier

```
python src/train_classifier.py
```

What it does: Takes the extracted features, selects the best 1,000 using statistical tests, balances classes with SMOTE, and trains an SVM with RBF kernel to classify Normal/Suspect/Keratoconus.

4. Evaluate

```
python src/evaluate.py
```

What it does: Tests the trained model on unseen data and calculates accuracy, precision, recall, F1-score, and AUC-ROC. Creates confusion matrices and ROC curves.

5. Generate Grad-CAM Visualizations

```
python src/gradcam_visualize.py
```

What it does: Creates heatmaps showing exactly which parts of the cornea image the AI focused on to make its decision. This helps doctors trust the AI.

Using Jupyter Notebooks (Interactive Learning)

Prefer clicking through code step by step? Use our notebooks:

```
# Start Jupyter
jupyter notebook

# Then open any .ipynb file in the notebooks/ folder
```

Notebook	What You' ll Learn
`01_data_exploration.ipynb`	What the images look like, class distribution
`02_single_map_baseline.ipynb`	Can one map alone detect keratoconus?
`03_multi_map_fusion.ipynb`	How combining all 7 maps improves results
`04_model_comparison.ipynb`	Why DenseNet-121 beat EfficientNet-B4 for us
`05_interpretability.ipynb`	How Grad-CAM explains AI decisions

Understanding the Output

Confusion Matrix

A table showing predictions vs. reality:

```
                Predicted
              Normal  Suspect  Keratoconus
Actual Normal     26        0            4
       Suspect     0       30            0
       KC          3        3           15
```

- **Good:** High numbers on the diagonal (correct predictions)
- **Problem:** Off-diagonal numbers show misclassifications

ROC Curve

A graph showing how well the model separates classes. The closer the curve is to the top-left corner, the better. **AUC = 0.94** means excellent separation.

Grad-CAM Heatmap

An image where:

- **Red/Yellow areas** = Where the AI looked to make its decision

- **Blue areas** = Ignored by the AI

For keratoconus, we expect red/yellow on the **inferior (lower) part of the cornea** — this is where the disease starts.

Common Issues & Fixes

Problem	Likely Cause	Solution
`ModuleNotFoundError`	Missing package	Run `pip install -r requirements.txt`
`CUDA out of memory`	GPU memory full	Reduce batch size in `config.py` or use CPU
`FileNotFoundError`	Dataset not in right place	Check folder structure matches Step 5
Results much worse than paper	Using wrong data split	Don' t mix train and test data
Slow training	No GPU	Normal on CPU; consider Google Colab for free GPU

Using Google Colab (Free GPU)

No GPU on your computer? Use Google's free servers:

1. Go to colab.research.google.com
2. Upload this repository
3. Change runtime to GPU: **Runtime → Change runtime type → GPU**
4. Run the notebook cells

For Non-Technical Readers

What is AI Doing Here?

Think of it like teaching a child to recognize apples:

1. You show thousands of apple pictures → "This is an apple"
2. The child learns patterns: round, red, has a stem
3. Later, you show a new picture → "Apple!"

Our AI does the same with corneas:

1. We show thousands of labeled eye maps → "This is keratoconus"
2. The AI learns patterns: steep curve, thin center, bulging back
3. Later, for a new patient → "Keratoconus — 87% confidence"

Why Should You Care?

- **For patients:** Earlier diagnosis means simpler treatment (cross-linking) instead of cornea transplant
- **For doctors:** AI assists, not replaces — catches cases that might be missed
- **For Bangladesh:** First step toward making specialist-level screening available everywhere

Is This Ready for Hospitals?

Not yet. This is research-level work. Before hospital use, we need:

- Larger Bangladeshi dataset (500+ patients)
- Regulatory approval (BMDC in Bangladesh)
- Real-time testing in clinical settings
- Integration with hospital software

But it's a strong foundation.

Citation

If you use this work in your research, please cite:

```
@thesis{azizul2026keratoconus,
  author     = {Azizul Hakim and Morshadul Islam Siam and A M Rohan},
  title      = {AI-Powered Multi-Map Framework for Keratoconus Classification in Bangladeshi Eye Images},
  school     = {Ahsanullah University of Science and Technology},
  year       = {2026},
  type       = {Undergraduate Thesis},
  supervisor = {Prof. Dr. Kazi A Kalpoma}
}
```

Contact & Collaboration

Role	Contact
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Clinical Partner	Dr. S M Enamul Haque —NIOH, Dhaka

Interested in collaboration? We're open to:

- Expanding the Bangladeshi dataset
- Testing on other South Asian populations
- Deploying in clinical settings
- Improving the model architecture

License

The **Bangladeshi dataset** is not publicly available due to patient privacy protections. Contact us for collaboration inquiries.

Acknowledgments

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- **AUST CSE Department** — for academic support and resources

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> *"Early detection saves sight. AI can help make that happen everywhere."*
> — Azizul Hakim, 2026

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